Year 10 Higher

# REVISION PACK

## Algebra 1

**Name:** _____

**Class:** _____

## Algebra 1 · Substitution and simplifying (Lessons 2.1, 2.2, 2.3)

**Worked example: substitution into v = u + at**

$v = u + at$. Work out the value of $v$ when $u = 12$, $a = -2$ and $t = 5$

| | | |
|---|---|---|
| **Step 1** | Put the numbers in | $v = 12 + (-2) \times 5$ |
| **Step 2** | Multiply first | $= 12 - 10$ |
| **Step 3** | Answer | $v = 2$ |

### Practice: substitution into v = u + at

**1.1** $v = u + at$. Work out the value of $v$ when $u = 3$, $a = -4$ and $t = 2$

**1.2** $v = u + at$. Work out the value of $v$ when $u = 10$, $a = -3$ and $t = 6$

**1.3** $v = u + at$. Work out the value of $v$ when $u = -4$, $a = 5$ and $t = 3$

**1.4** $v = u + at$. Work out the value of $v$ when $u = 20$, $a = -6$ and $t = 3$

**1.5** $v = u + at$. Work out the value of $v$ when $u = -2$, $a = -3$ and $t = 4$

**1.6** $v = u + at$. Work out the value of $v$ when $u = 15$, $a = -2.5$ and $t = 4$

## Algebra 1 · Substitution and simplifying (Lessons 2.1, 2.2, 2.3)

**Worked example: substitution with a square**

$E = \frac{1}{2}mv^2$. Work out the value of $E$ when $m = 2$ and $v = -7$

| | | |
|---|---|---|
| **Step 1** | Put the numbers in | $E = \frac{1}{2} \times 2 \times (-7)^2$ |
| **Step 2** | Square first | $= \frac{1}{2} \times 2 \times 49$ |
| **Step 3** | Answer | $E = 49$ |

### Practice: substitution with a square

**1.7** $E = \frac{1}{2}mv^2$. Work out the value of $E$ when $m = 6$ and $v = -5$

**1.8** $E = \frac{1}{2}mv^2$. Work out the value of $E$ when $m = 10$ and $v = -2$

**1.9** $E = \frac{1}{2}mv^2$. Work out the value of $E$ when $m = 8$ and $v = -3$

**1.10** $E = \frac{1}{2}mv^2$. Work out the value of $E$ when $m = 4$ and $v = -6$

**1.11** $E = \frac{1}{2}mv^2$. Work out the value of $E$ when $m = 12$ and $v = -1$

**1.12** $E = \frac{1}{2}mv^2$. Work out the value of $E$ when $m = 3$ and $v = -4$

## Algebra 1 · Substitution and simplifying (Lessons 2.1, 2.2, 2.3)

**Worked example: simplifying expressions**

Simplify

(a) $6p + 2q - 4p + 3q$ (b) $2a^2b \times 5ab^3$

| | | | |
|---|---|---|---|
| **(a)** | **Step 1** | Collect $p$ | $6p - 4p = 2p$ |
| | **Step 2** | Collect $q$ | $2q + 3q = 5q$ |
| | **Step 3** | Answer | $2p + 5q$ |
| **(b)** | **Step 1** | Multiply the numbers | $2 \times 5 = 10$ |
| | **Step 2** | Add the powers | $a^{2+1}b^{1+3} = a^3b^4$ |
| | **Step 3** | Answer | $10a^3b^4$ |

### Practice: simplifying expressions

**1.13** Simplify

(a) $7x + 4y - 3x - y$ (b) $4a^3b \times 2ab^2$

**1.14** Simplify

(a) $5m - 2n + m + 6n$ (b) $3xy^2 \times 6x^2y$

**1.15** Simplify

(a) $9c + 3d - 4c - 7d$ (b) $5p^2q^2 \times 2pq^4$

**1.16** Simplify

(a) $8x - 3y + 2x + 7y$ (b) $6a^2b^3 \times 3ab$

**1.17** Simplify

(a) $4m + 9n - 7m - 2n$ (b) $2x^3y \times 7x^2y^2$

**1.18** Simplify

(a) $3p - 5q + 4p + 8q$ (b) $5ab^2 \times 4a^3b$

**Worked example: unknowns on both sides**

Solve $7x - 4 = 3x + 16$

| | |
|---|---|
| **Step 1** Subtract $3x$ | $4x - 4 = 16$ |
| **Step 2** Add 4 | $4x = 20$ |
| **Step 3** Divide by 4 | $x = 5$ |

## Practice: unknowns on both sides

**2.1** Solve $6x - 5 = 2x + 11$

**2.2** Solve $9x + 4 = 5x + 24$

**2.3** Solve $8x - 3 = 5x + 12$

**2.4** Solve $7x + 2 = 3x - 10$

**2.5** Solve $10x - 9 = 4x + 15$

**2.6** Solve $5x + 8 = 2x + 23$

## Algebra 1 · Solving equations (Lesson 2.4)

**Worked example: equations with fractions**

Solve $\dfrac{3x-1}{4}=5$

| | | |
|---|---|---|
| **Step 1** | Multiply by 4 | $3x-1=20$ |
| **Step 2** | Add 1 | $3x=21$ |
| **Step 3** | Divide by 3 | $x=7$ |

### Practice: equations with fractions

**2.7** Solve $\dfrac{2x+5}{3}=7$

**2.8** Solve $\dfrac{4x-3}{5}=5$

**2.9** Solve $\dfrac{5x+2}{2}=6$

**2.10** Solve $\dfrac{3x+4}{5}=2$

**2.11** Solve $\dfrac{2x-1}{3}=5$

**2.12** Solve $\dfrac{6x+3}{3}=5$

**Worked example: equations with brackets**

Solve $2(3x+1)=5(x+2)$

| | | |
|---|---|---|
| **Step 1** | Expand | $6x+2=5x+10$ |
| **Step 2** | Subtract $5x$ | $x+2=10$ |
| **Step 3** | Subtract 2 | $x=8$ |

## Practice: equations with brackets

**2.13** Solve $4(2x-1)=3(x+7)$

**2.14** Solve $5(x-2)=2(2x+3)$

**2.15** Solve $3(3x+2)=4(x+9)$

**2.16** Solve $2(4x-3)=3(x+3)$

**2.17** Solve $6(x-1)=4(x+2)$

**2.18** Solve $4(2x+1)=2(3x+1)$